

Damage Grade Assessment - LAMJUNG

13 FEB 2022

Presented by: HRRP



www.hrrpnepal.org

Funded by



facebook.com/HRRPNepal

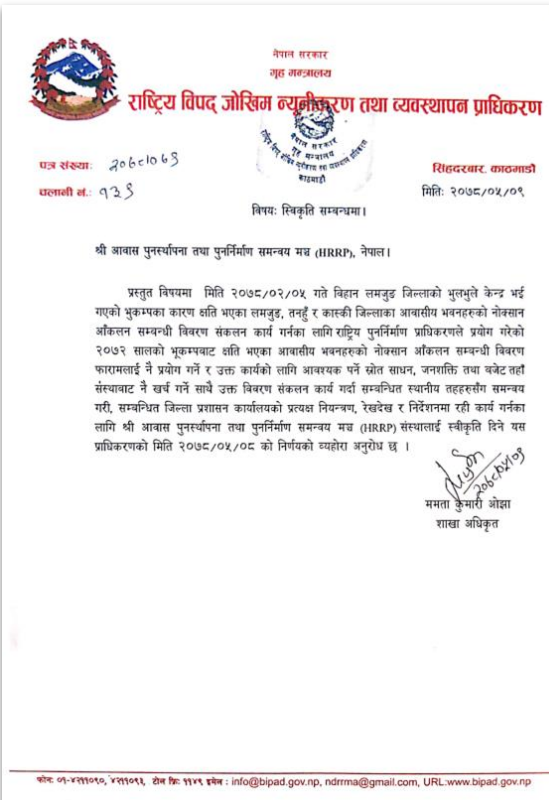
AGENDA



- Background & Methodology
- Key findings of Damage Grade Assessment of Lamjung district
 - Status of survey
 - DG - Analysis
- Limitation - sample size
- Recommendations

BACKGROUND

- On 19 May 2021, Wednesday at 5:42 AM, the earthquake measuring 5.8M followed by subsequent aftershocks hit Lamjung district of Gandaki Province in Nepal.
- Preliminary information from the DDMC Lamjung, 1046 HHs are fully damaged and 1465 HHs are partially damage by the earthquake. In today 2521 HHs are affected by the earthquake.
- District Disaster Management Committee (DDMC) with the technical support from HRRP has started DGA at field level from 10th Sep 2021.
- 6 municipalities were affected from the earthquake : Dordi, Besisahar and Marsyangdi are highly affected and Kholasothar and Dudhpokhari are less affected.



नेपाल सरकार
गृह मन्त्रालय

राष्ट्रिय विपद् जोखिम न्यूनीकरण तथा व्यवस्थापन प्राधिकरण

पत्र संख्या: २०६८०६९
चलानी सं: १३९

सिंहदरबार, काठमाडौं
मिति: २०७८/०५/०९

विषय: स्थिति सम्बन्धमा।

श्री आवास पुनर्स्थापना तथा पुनर्निर्माण समन्वय मण्डल (HRRP), नेपाल।

प्रस्तुत विषयमा मिति २०७८/०२/०५ गते बिहान सम्बन्धित जिल्लाको भुतभुते केन्द्र भई गएको भूकम्पका कारण क्षति भएका ताम्बुङ, तनहुँ र कास्की जिल्लाका आवासीय भवनहरूको नोक्सान अंकलन सम्बन्धी विवरण संकलन कार्य गर्नका लागि राष्ट्रिय पुनर्निर्माण प्राधिकरणले प्रयोग गरेको २०७२ सालको भूकम्पबाट क्षति भएका आवासीय भवनहरूको नोक्सान अंकलन सम्बन्धी विवरण फारमलाई नै प्रयोग गर्ने र उक्त कार्यको लागि आवश्यक पर्ने खेत साधन, जनशक्ति तथा बजेट तहोता संस्थाबाट नै खर्च गर्ने साथै उक्त विवरण संकलन कार्य गर्दा सम्बन्धित स्थानीय तहहरूसँग समन्वय गरी, सम्बन्धित जिल्ला प्रशासन कार्यालयको प्रत्यक्ष नियन्त्रण, रेखदेख र निरीक्षणमा रही कार्य गर्नका लागि श्री आवास पुनर्स्थापना तथा पुनर्निर्माण समन्वय मण्डल (HRRP) संस्थालाई स्वीकृति दिने यस प्राधिकरणको मिति २०७८/०५/०८ को निर्णयको व्युत्पन्न अनुरोध छ।

(Signature)
ममता कुमारी ओझा
शाखा अधिकृत

फोन: ०१-४२२१०१६, ४२१०११, टोल फ्री: ११९९ ईमेल: info@bipad.gov.np, ndrma@gmail.com, URL: www.bipad.gov.np

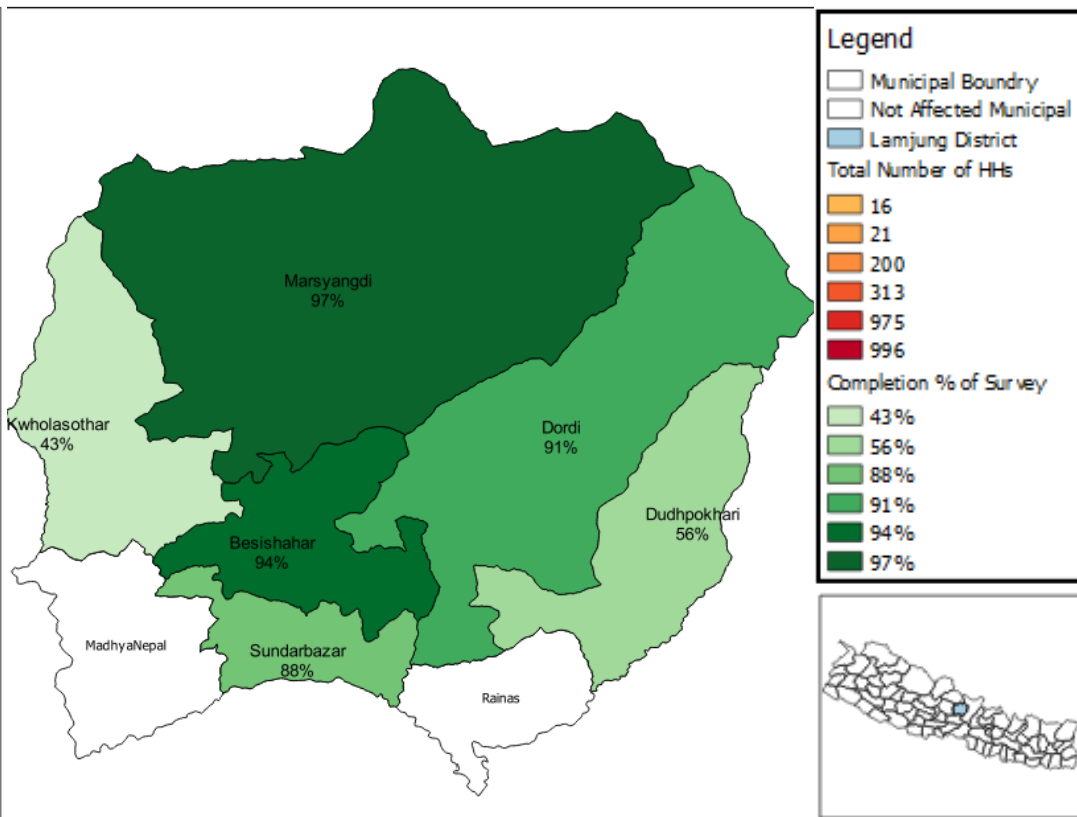
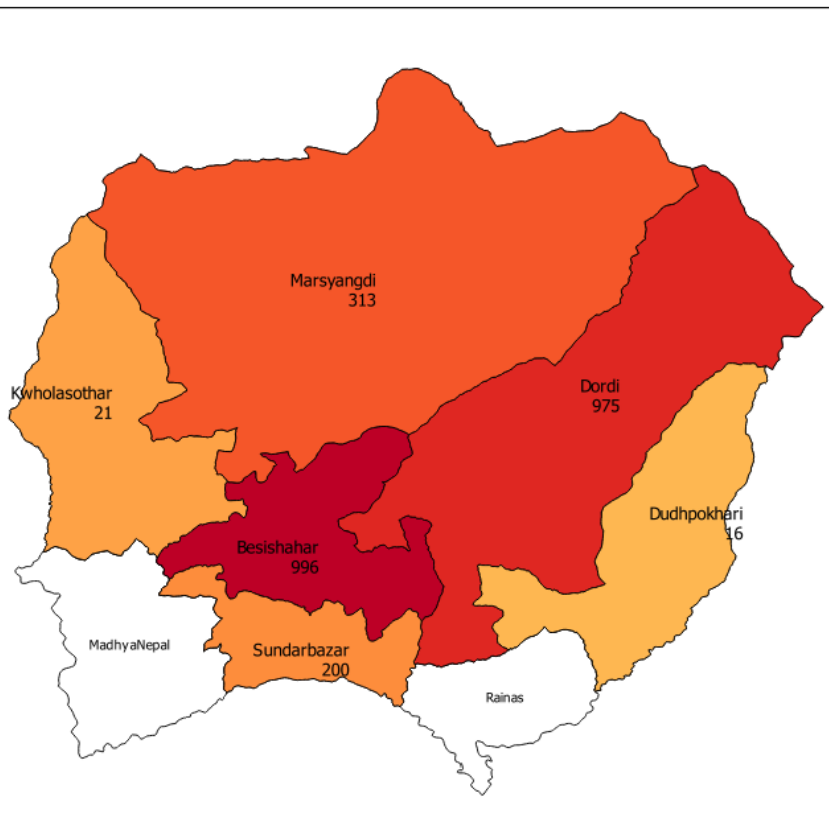
METHODOLOGY



- Conducted 3 days orientation session for 14 technical staff (engineers and sub engineers) at field level.
- 14 NRA local technical staffs are temporarily hired for the assessment.
- Adapted same questionnaire from 2015 CBS Damage Grade Assessment; questionnaire developed and verified by NDRRMA and NRA.
- Quantitative data collection method is applied where technical staff have visited all the affected households and collect necessary data/information.
- Coordination and collaboration with district, municipalities and ward representatives were done through out the assessment.
- HRRP technical staff supported to cover up the GAP areas(Delay in assessment)



OVERVIEW



STATUS OF SURVEY



2,429 Total House Holds



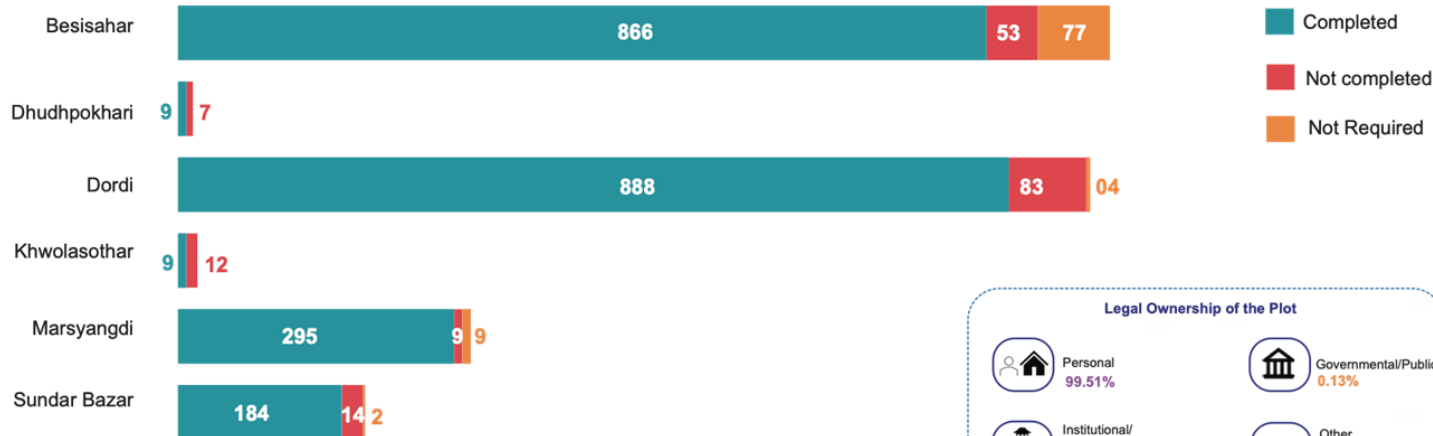
92 HHS are Duplicate entry, community Building, School, Temple etc. in the list of **2521**



2,251 Total House Holds Surveyed



178 HHs not found/ were not available



Legal Ownership of the Plot



Personal
99.51%



Governmental/Public
0.13%

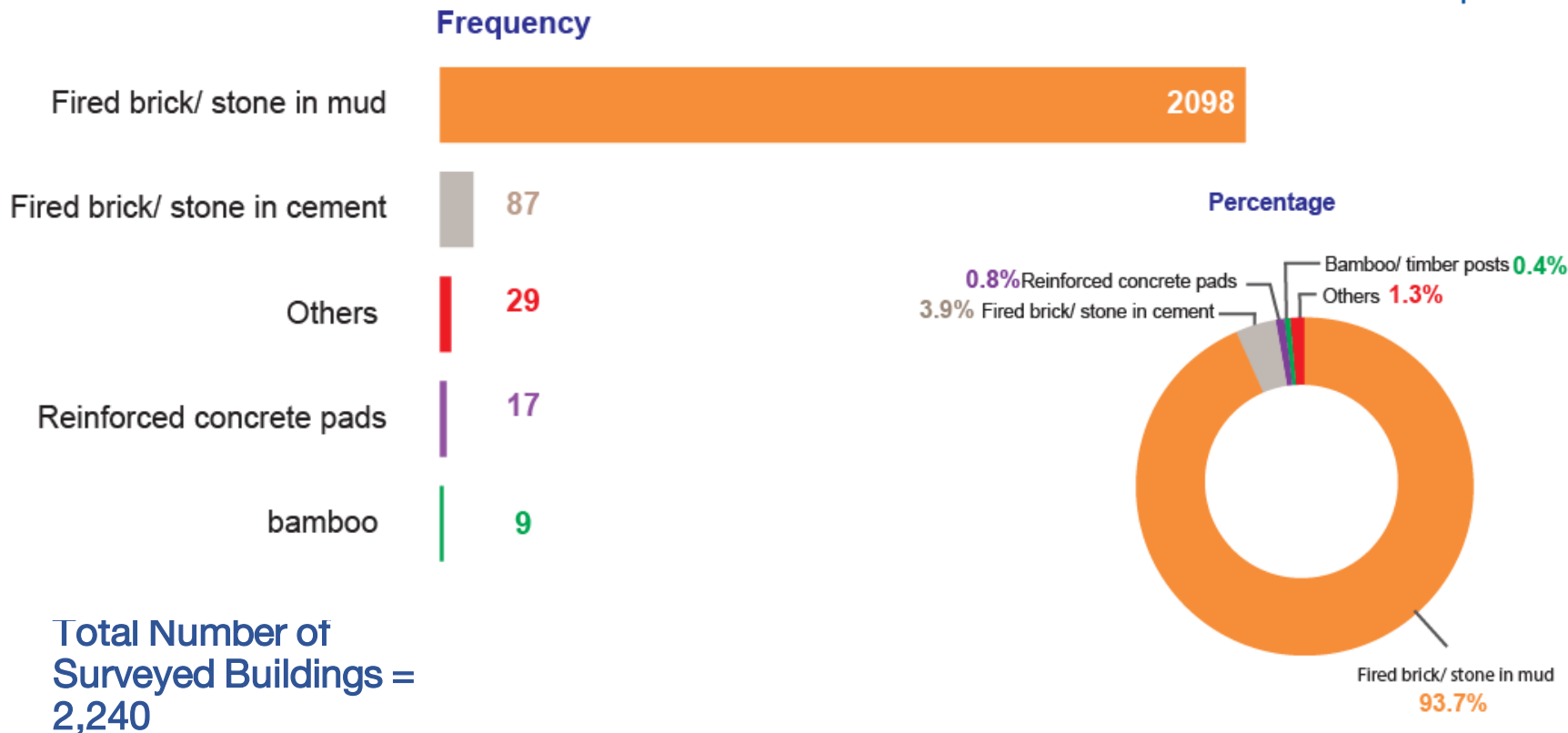


Institutional/
Community(Guthi)
0.22%



Other
0.13%

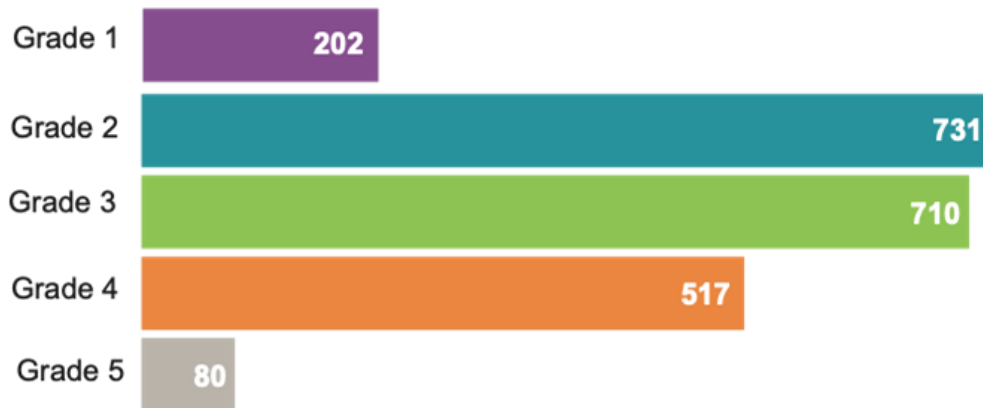
DISTRIBUTION OF BUILDING TYPOLOGY



DAMAGED GRADE STATISTIC

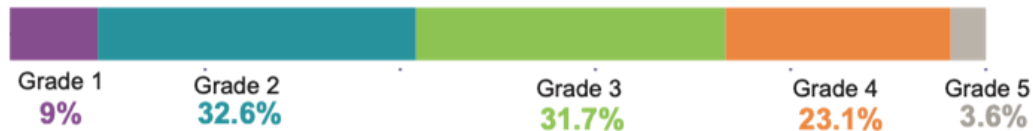


Frequency

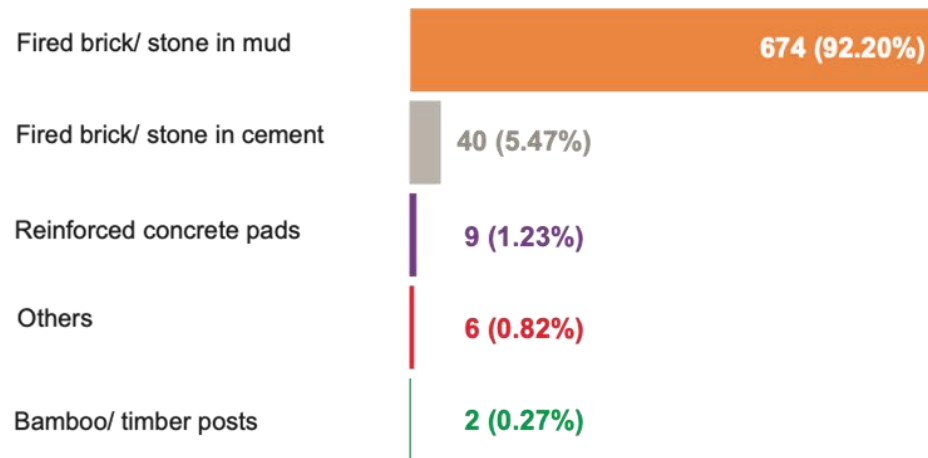
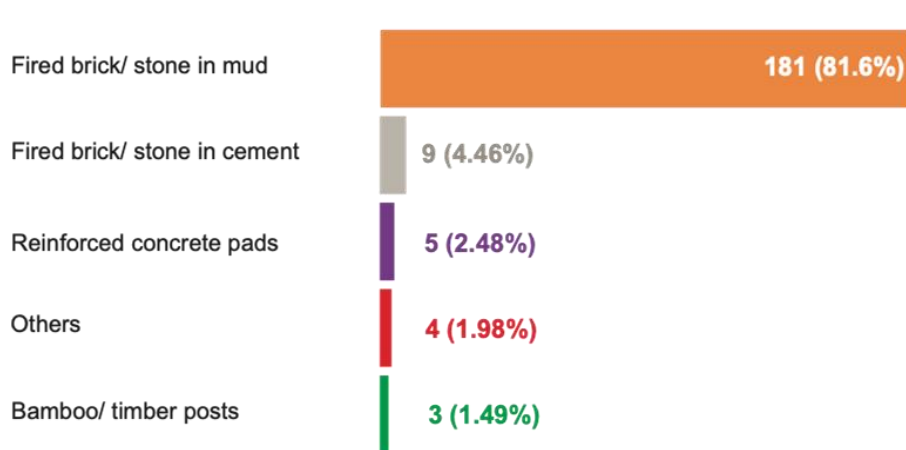


Total Number of
Surveyed Buildings =
2,240

Percentage



BUILDING TYPOLOGY AND DAMAGED GRADE - DG1 & DG2



Damage Grade 1 Vs Typology

Damage Grade 2 Vs
Typology

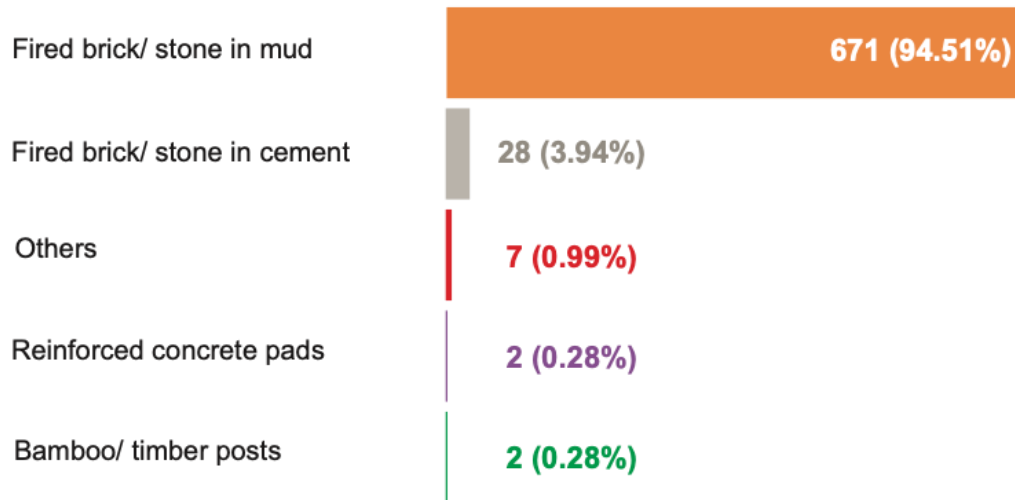
DAMAGED GRADE- DG1 BUILDINGS EXAMPLES



DAMAGED GRADE- DG2 BUILDINGS EXAMPLES



DAMAGE GRADE, DG3 Vs Typology



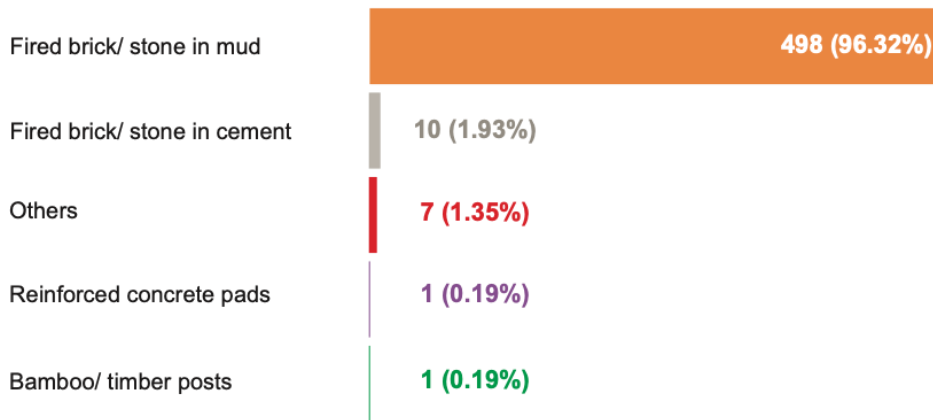
Damage Grade 3 Vs Typology



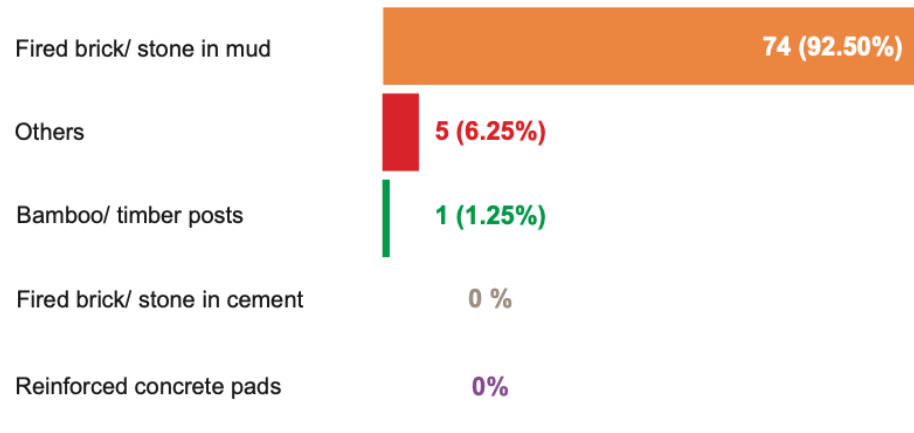
DAMAGED GRADE- DG3 BUILDINGS EXAMPLES



DAMAGE GRADE 4 & 5 Vs Typology



Damage Grade 4



Damage Grade 5

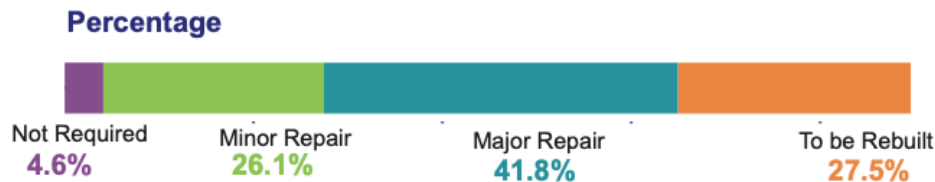
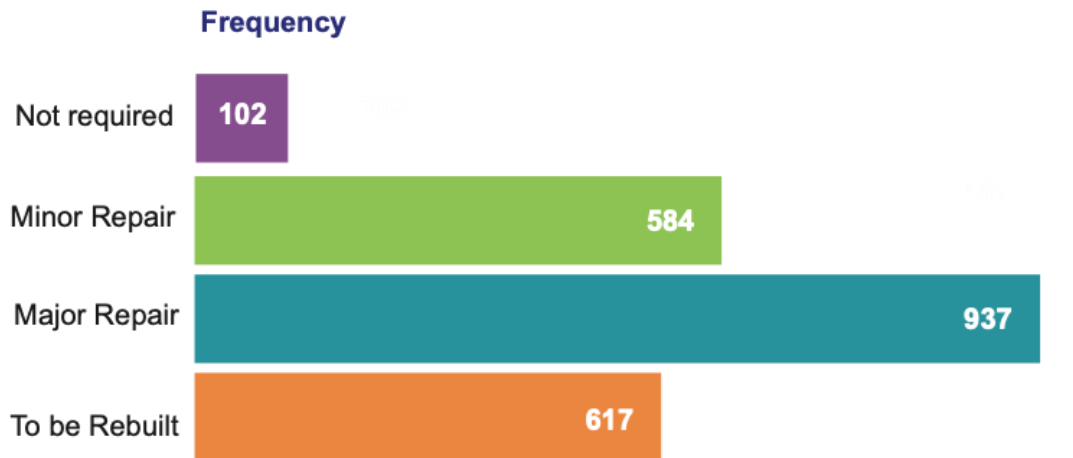
DAMAGED GRADE- DG4 BUILDINGS EXAMPLES



DAMAGED GRADE- DG5 BUILDINGS EXAMPLES



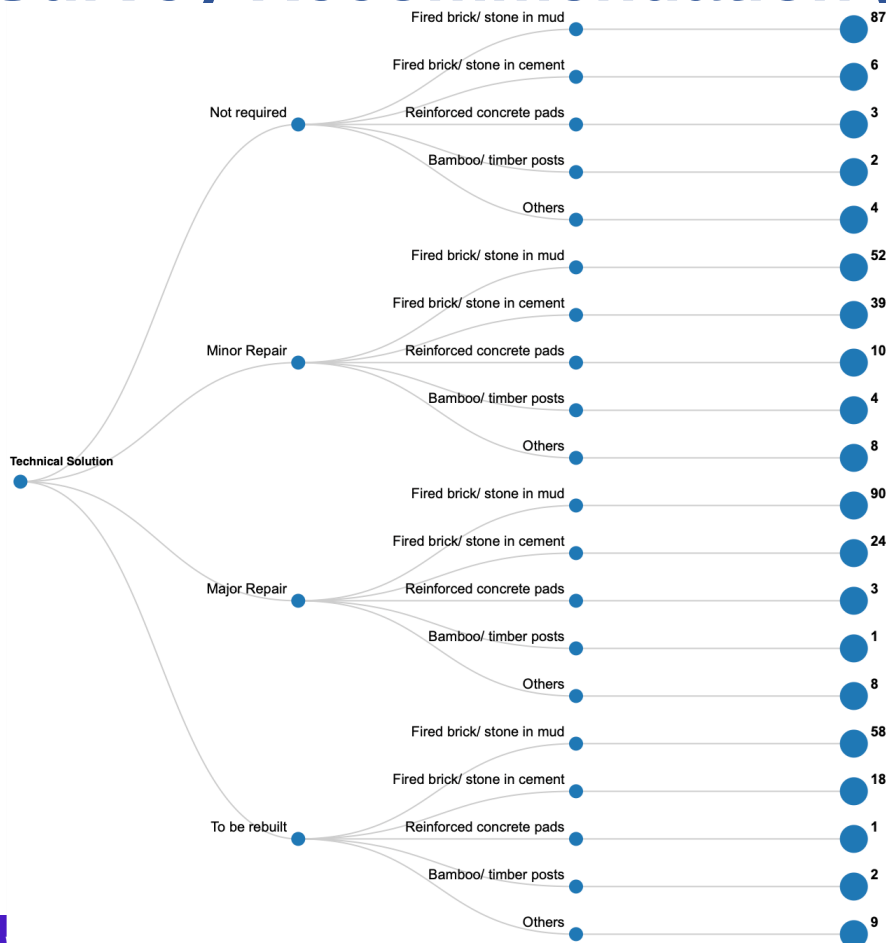
Survey Recommendation(Overall Status)



Total Number of
Surveyed Buildings
=2,240

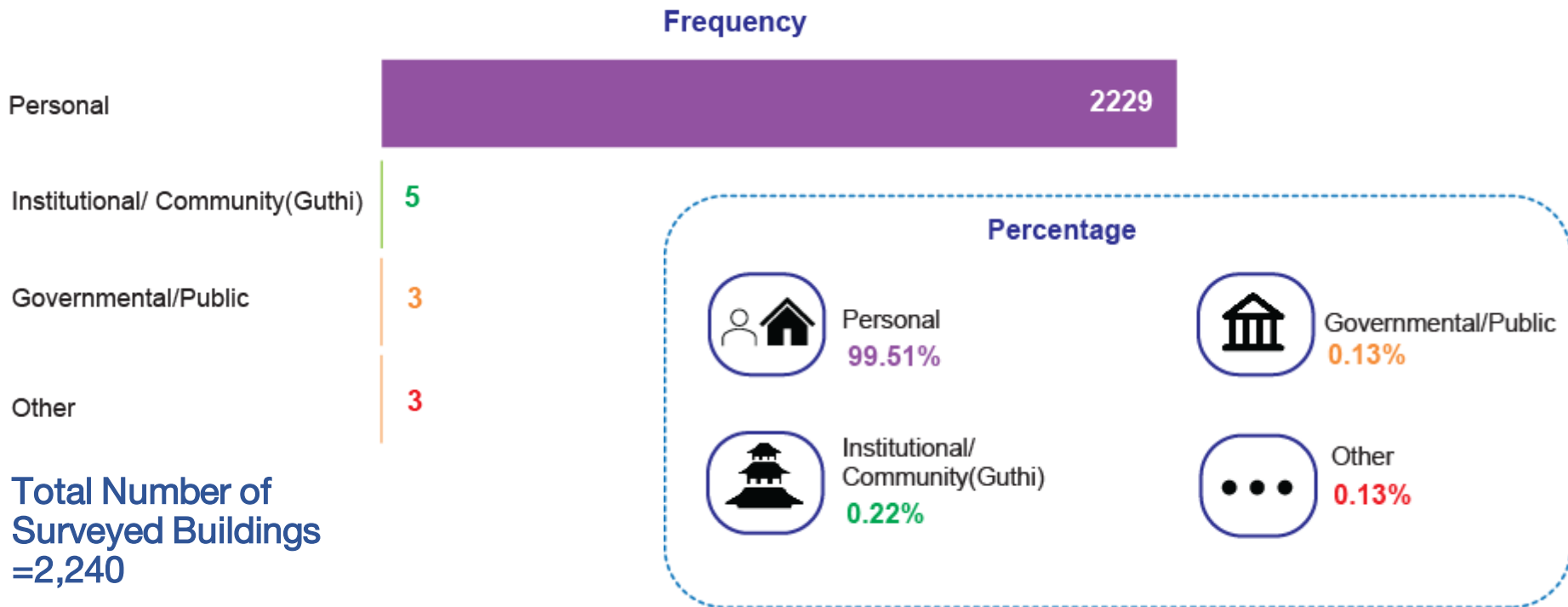
Type of Technical Solution

Survey Recommendation (Building Typology)

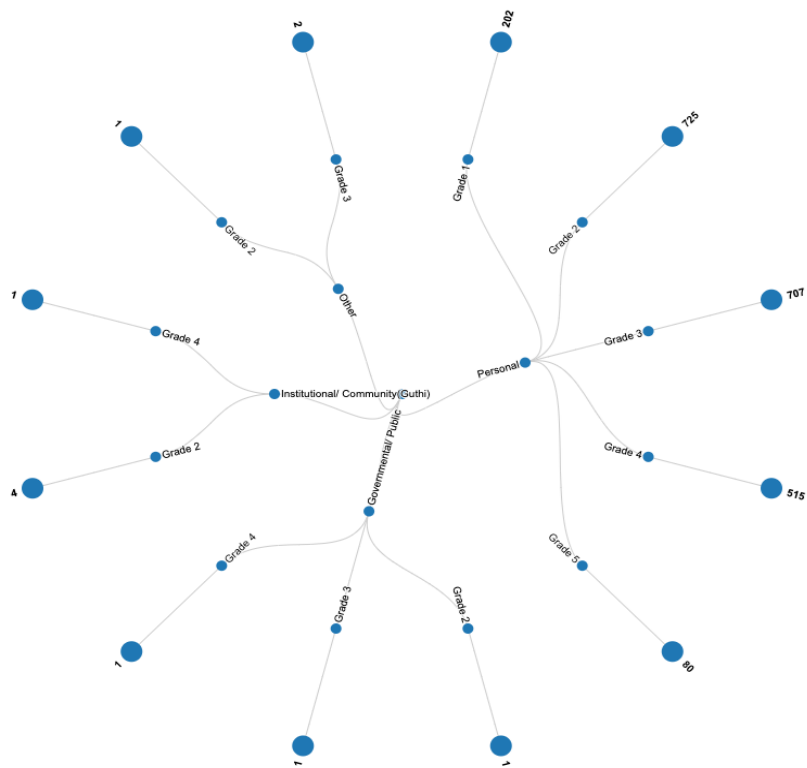


Technical Solution		Building Typology					Total
		Fired brick/ stone in mud	Fired brick/ stone in cement	Reinforced concrete pads	Bamboo/ timber posts	Others	
Not required	Count	87	6	3	2	4	102
	%	85.3%	5.9%	2.9%	2.0%	3.9%	100.0%
Minor Repair	Count	523	39	10	4	8	584
	%	89.6%	6.7%	1.7%	0.7%	1.4%	100.0%
Major Repair	Count	901	24	3	1	8	937
	%	96.2%	2.6%	0.3%	0.1%	0.9%	100.0%
To be rebuilt	Count	587	18	1	2	9	617
	%	95.1%	2.9%	0.2%	0.3%	1.5%	100.0%
Total	Count	2098	87	17	9	29	2240
	%	93.7%	3.9%	0.8%	0.4%	1.3%	100.0%

LEGAL OWNERSHIP STATUS of the PLOT of LAND of BUILDING BUILT



Damage Grade Vs Legal Ownership

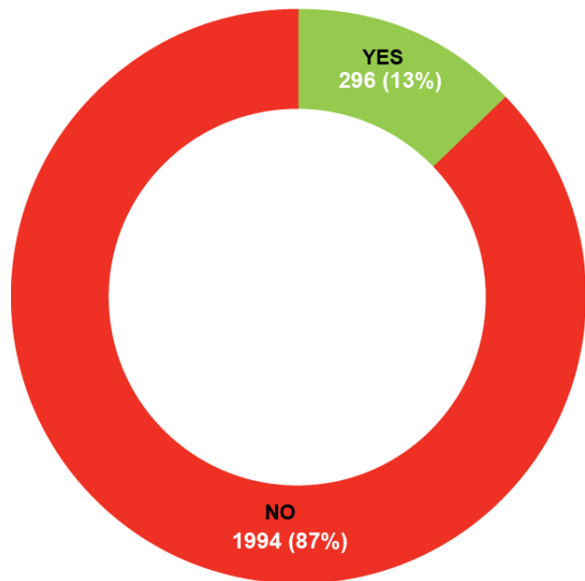


Damage grade		The legal ownership status of the plot of land of the house				Total
		Personal	Institutional/ Community (Guthi)	Governmental/ Public	Other	
Grade 1	Count	202	0	0	0	202
	%	100.0%	0.0%	0.0%	0.0%	100.0%
Grade 2	Count	725	4	1	1	731
	%	99.2%	0.5%	0.1%	0.1%	100.0%
Grade 3	Count	707	0	1	2	710
	%	99.6%	0.0%	0.1%	0.3%	100.0%
Grade 4	Count	515	1	1	0	517
	%	99.6%	0.2%	0.2%	0.0%	100.0%
Grade 5	Count	80	0	0	0	80
	%	100.0%	0.0%	0.0%	0.0%	100.0%

STATUS of TECHNICAL SOLUTION PROVIDED Vs LEGAL OWNERSHIP

Technical Solution		The legal ownership status of the plot of land of the house				Total
		Personal	Institutional/ Community(Guthi)	Governmental/ Public	Other	
Not required	Count	102	0	0	0	102
	%	100.0%	0.0%	0.0%	0.0%	100.0%
Minor Repair	Count	579	2	2	1	584
	%	99.1%	0.3%	0.3%	0.2%	100.0%
Major Repair	Count	934	2	0	1	937
	%	99.7%	0.2%	0.0%	0.1%	100.0%
To be rebuilt	Count	614	1	1	1	617
	%	99.5%	0.2%	0.2%	0.2%	100.0%
Total	Count	2229	5	3	3	2240
	%	99.5%	.2%	.1%	.1%	100.0%

GEOTECHNICAL RISK OF THE AREA



Landslide

35%

Ground Fissures

23.9%

Flood

15.3%

Rock Fall

10.2%

Liquefaction

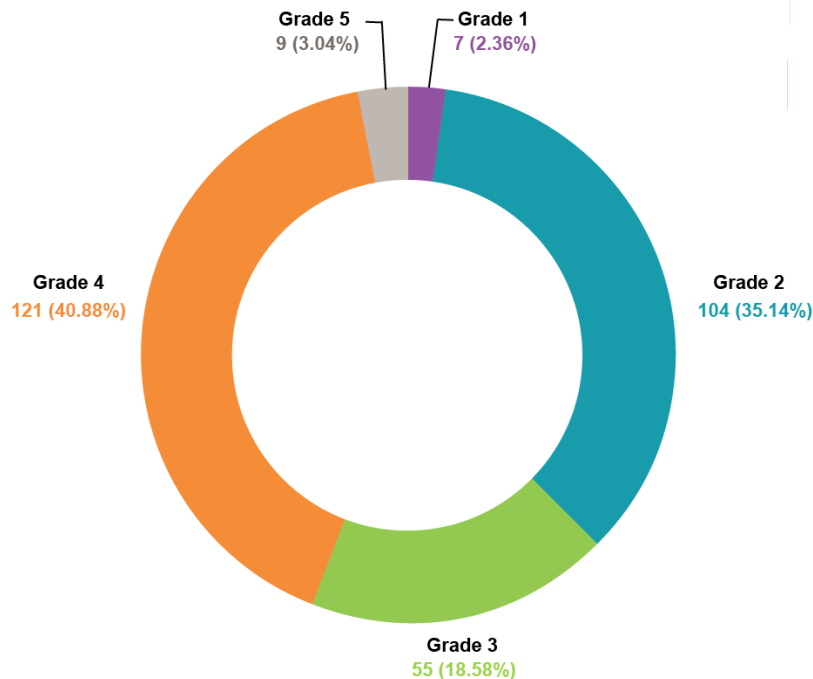
9%

Ground Displacement

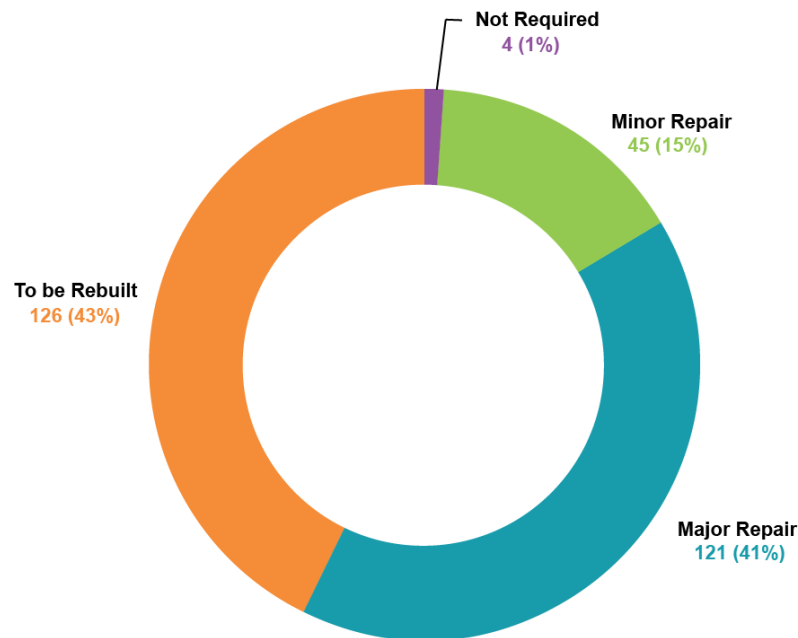
6.6%

Landslides, Ground Fissures and Flood were the major type of geotechnical risks faced in the area where buildings were built.

DAMAGE GRADE Vs TECHNICAL SOLUTION OF HOUSES in GEOTECHNICAL RISK AREA



Damage Grade of House in Risk Area



Technical Solutions

SURVEY LIMITATION



- Survey design for 2521 HHs as per the existing damage data provided by the DDMC.
- Use available survey form designed for Gorkha Earthquake Housing reconstruction.

RECOMMENDATION



- **From Survey Analysis**
 - DG I : NA
 - DG II and DG III: Retrofitting
 - DGIV and DGV : Reconstruction
- **Case load according to the above recommendations:**
 - DGI: 202 HHs=Non beneficiaries; Gov. Grant Not recommended
 - DGII and DGIII: 1441 HHs = Retrofitting Beneficiaries ; Gov. Grant Recommended
 - DGIV and DGV: 597 HHs = New Construction Beneficiaries (Full Beneficiaries); Gov. Grant Recommended
- **Government Retrofitting support grant amount revision**

Retrofitting Cost in average: NPR 300K-400K for stone masonry

New construction Cost in average: NPR 600K for stone masonry with mud

 - NRA grant for retrofitting is NPR 100K and New construction Grant NPR 300K
 - Recommend **NDRRMA to increase the retrofitting grant** to promote retrofitting and save investment on reconstruction achieving increased seismic performance of damaged building.

BUDGET SUMMARY AS PER RECOMMENDATION

Beneficiaries Type	Case load as per Recommendation (HHs)	Gov.Grant similar to NRA provision (NRs)	Total Budget (NRs)	Gov. Grant support (NRs)	Total Budget (NRs)	Budget Difference (NRs)
Retrofitting (DG2 & DG3)	1441	100,000	144,100,000	200,000	288,200,000	(144,100,000)
New Re-construction (DG4 & DG5)	597	300,000	179,100,000	400,000	238,800,000	(59,700,000)
Total	2,038	400,000	323,200,000	600,000	527,000,000	(203,800,000)

NDRRMA current provision: Retrofitting cost is 50K/Beneficiaries, Total budget= Retrofitting (NRs 72,050,000) + New construction (238,800,000)=**NRs. 310,850,000**

Alt. OPTIONS-1

Beneficiaries Type	Case load as per Recommendation (HHs)	Gov.Grant similar to NRA provision (NRs)	Total Budget (NRs)	Gov. Grant support (NRs)	Total Budget (NRs)	Budget Difference (NRs)
Retrofitting (DG1 & DG2)	933	100,000	93,300,000	200,000	186,600,000	(93,300,000)
New Re-construction (DG3, DG4 & DG5)	1307	300,000	392,100,000	400,000	522,800,000	(130,700,000)
Total	2,240		485,400,000		709,400,000	(224,000,000)

NDRRMA current provision: Retrofitting cost is 50K/Beneficiaries, Total budget: Retrofitting (NRs 46,650,000) + New construction (522,800,000) = **NRs. 569,450,000**

Alt options-2

Beneficiaries Type	Case load as per Recommendation (HHs)	Gov.Grant similar to NRA provision (NRs)	Total Budget (NRs)	Gov. Grant support (NRs)	Total Budget (NRs)	Budget Difference (NRs)
Retrofitting (DG2)	731	100,000	73,100,000	200,000	146,200,000	(73,100,000)
New Re-construction (DG3, DG4 & DG5)	1307	300,000	392,100,000	400,000	522,800,000	(130,700,000)
Total	2,038		465,200,000		669,000,000	(203,800,000)

NDRRMA current provision: Retrofitting cost is 50K/Beneficiaries, Total budget= Retrofitting (NRs 36,550,000) + New construction (522,800,000)=**NRs. 559,350,000**

Alt. options-3

Beneficiaries Type	Case load as per Recommendation (HHs)	Gov.Grant similar to NRA provision (NRs)	Total Budget (NRs)	Gov. Grant support (NRs)	Total Budget (NRs)	Budget Difference (NRs)
Retrofitting (DG1, DG2 & DG3)	1643	100,000	164,300,000	200,000	328,600,000	(164,300,000)
New Re-construction (DG4 & DG5)	597	300,000	179,100,000	400,000	238,800,000	(59,700,000)
Total	2,240		343,400,000		567,400,000	(224,000,000)

NDRRMA current provision: Retrofitting cost is 50K/Beneficiaries, Total budget= Retrofitting (NRs 82,150,000) + New construction (238,800,000)=**NRs. 320,950,000**

WAY FORWARDS

1. Develop full assessment report Jointly with NDRRMA technical team
2. Assign focal person from NDRRMA for Lamjung Earthquake Recovery
3. Prepare for grievance collection and addressing mechanisms
4. Process for required decision and endorsement
5. Announcement of Government Subsidy (4 Lakh for reconstruction and retrofit)
6. Launch Post Earthquake Recovery and Reconstruction Plan adapting 2015 PDRF.
7. Enrollment Program and Announcement event at district level.
8. NHSRP(HRRP) to coordinate STA with other partners

THANK-YOU



www.hrrpnepal.org

Funded by



facebook.com/HRRPNepal